1 Introduction to Signalling Cables

The Function of Cables

Introduction to this Module

This learning module provides the basic theoretical knowledge of signalling cables. You will need to understand this before progressing to the modules which will give you the practical skills you will need as a signalling technician. It describes the construction, inspection and testing methods used to prove the integrity of signalling cables. By following the assignments and self assessment exercises you will gain the required knowledge to support the further practical training and assessment you will receive before your authorisation to carry out these tasks in the workplace.

Signalling Circuits

The signalling circuits you will have to maintain and fault comprise a number of components which need to be connected electrically. In most cases the connections have to be made in a particular way to achieve important functions such as the interlocking of controls.

Use of Cables for Signalling

The cables are used to connect the components in signalling circuits and are, therefore, a vital part of the interlocking system. They have two main functions:

- **communication** - passing controls from the interlocking to trackside equipment and indications of equipment state back to the signal box.
- **power supply** - carrying electrical power to signalling circuit components.

1.2 Cable Construction

Component Parts

A **cable** is a wire or group of wires each with sufficient **insulation** to isolate it electrically and to protect it from damage. A cable has four main components

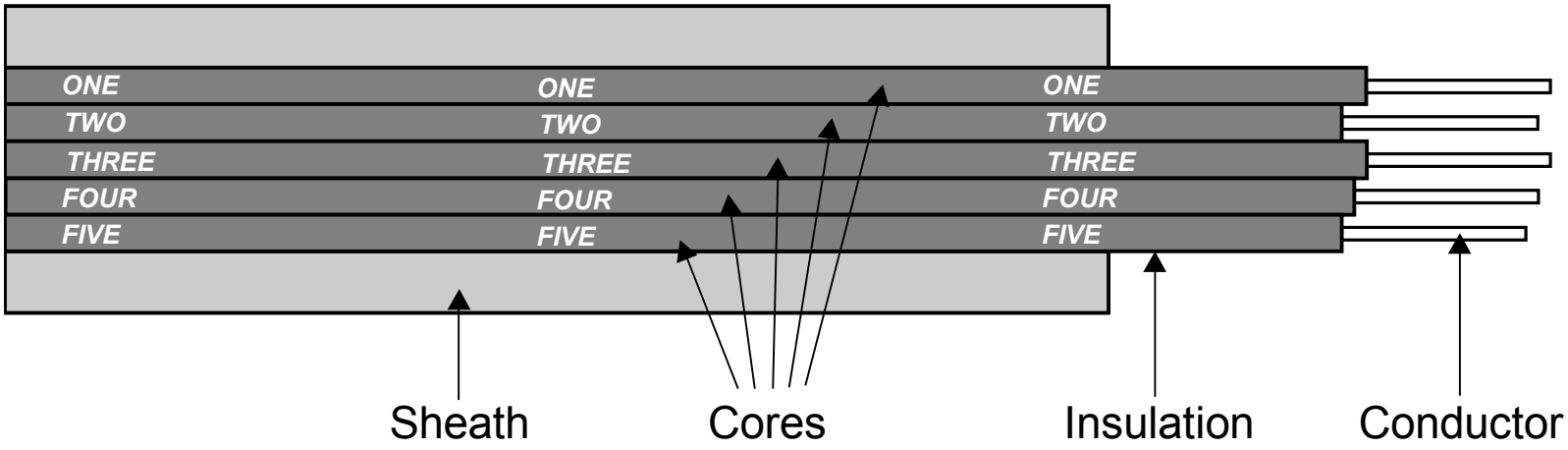
- **conductor**
- **core**
- **insulation**
- **sheath**

The Conductor

The **conductor** carries the electrical current and is usually made of *copper*. Some power supply conductors are made of *aluminium* which can be more cost effective. The size of the conductor limits the current carrying capacity of the cable, i.e. the larger the conductor, the lower the resistance and, consequently, the larger the current carrying capacity. If a short circuit occurs, the current may be very high for a short period before the fuse blows.

Example of Cable Construction

The resulting heat must not be allowed to damage the cable, so the conductor is made large enough to withstand this current surge without overheating. Where the requirement is for a large current carrying capacity the conductor is sometimes made up of many strands of thin wire rather than one large, solid conductor. This makes the cable more flexible, easier to handle and less vulnerable to stress damage.



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Figure 1

The Core

A **core** is a single conductor in a group of conductors but insulated from all others in what is called a multi-core cable. In a **multi-core** cable cores are numbered from the centre of the cable outwards. Number 1 is always at the centre and the end of the cable can be identified by the rotation of the numbering.

- **'A' end** is the end of the cable where the numbering of the cores increases clockwise from the centre. The A end of a cable is shown in figure 2.
- **'Z' end** is the end of the cable where the numbering of the cores increases anti-clockwise from the centre of the cable. The Z end of a cable is shown in figure 3.

Insulation

The **insulation** keeps each conductor electrically separate from other conductors in the cable. Also, it protects the conductor from damage. The type of insulation and its thickness will depend on the circuit voltage and the use of the cable.

The Sheath

The outer layer of a cable is known as the **sheath**. It is designed to protect the cable from damage and deterioration. In some cases, depending on its use, the sheath may include metallic armour. The sheath also increases the overall insulation resistance of the cable.

Note: Both insulation and sheathing have the following characteristics

- resistance to heat, moisture and physical damage.
- flexibility.

‘A’ End and ‘Z’ End of a Multi-core Cable

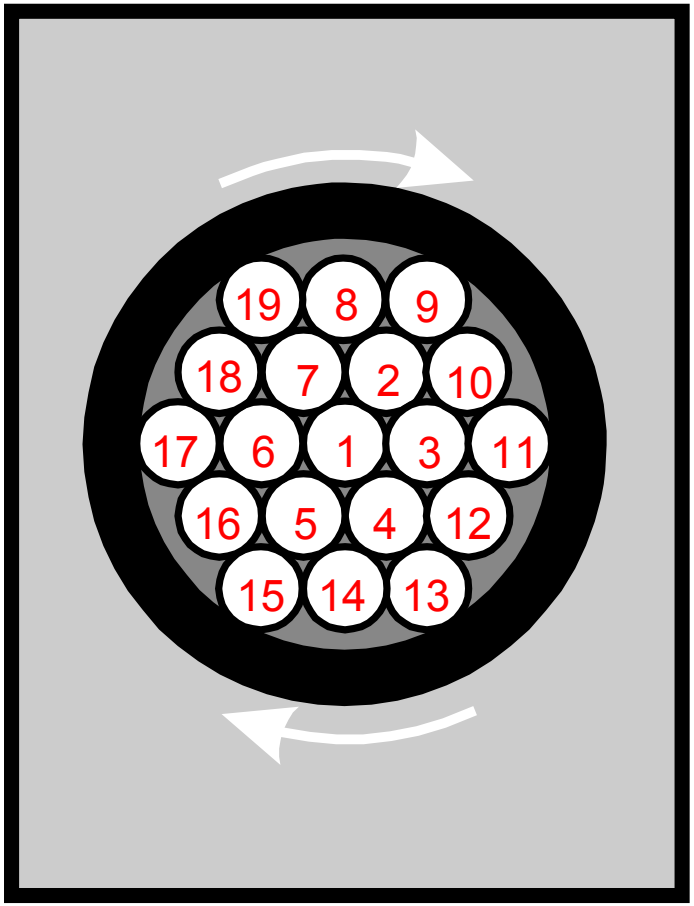


Figure 2. ‘A’ End of a Multicore cable

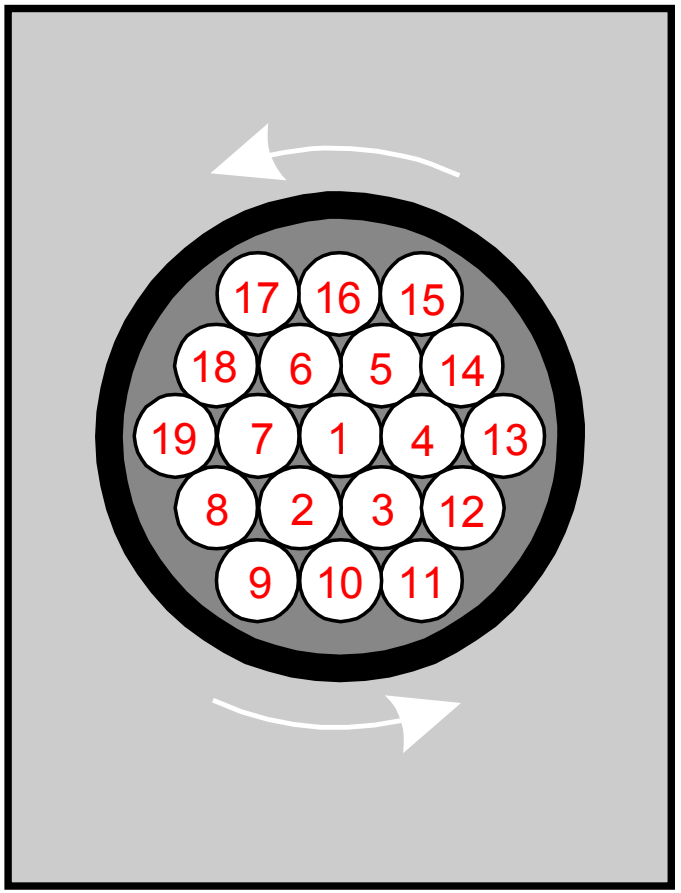


Figure 3. ‘Z’ End of Multicore cable

1.3 Types of Signalling Cable

Cables are used for communication between the signalbox and trackside equipment. The communication is two way.

- Control instructions are passed to operate the trackside equipment.
- Information is sent back from the trackside to the signal box on a separate conductor.

Other cables transmit the electric power to operate trackside equipment.

The three types of cable which carry out these functions are

- lineside cables
- tail cables
- power cables

The use of these cables in a signalling system is shown in figure 4.

Use of Cables in the Signalling System

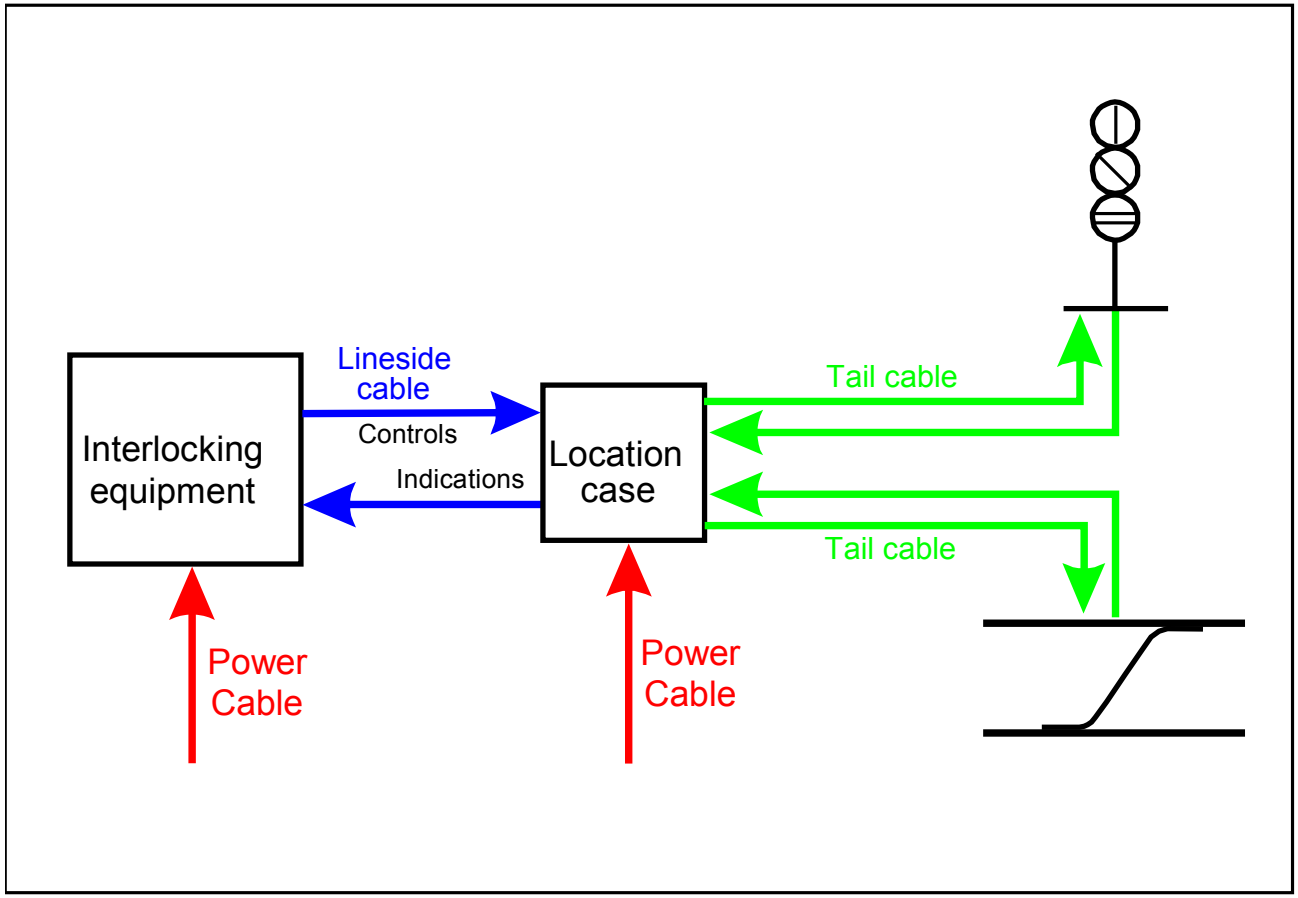


Figure 4

Lineside Cables

Lineside cables run from the signalbox to location cases and between cases. They may be above ground, in concrete troughing, or buried. A cable can have up to 48 cores and carries a number of different commands or indications for a variety of equipment. A photograph showing a lineside cable is at figure 5.

Not all of the conductors carry information. In each cable there are usually a number of spares. These can be used to

- replace faulty conductors without the necessity of replacing the complete cable
- allow connection of additional equipment at a later date

Tail Cables

Tail cables run from location cases to trackside equipment. They are usually above ground and are routed under tracks through orange piping. A photograph showing a tail cable is at figure 6.

A tail cable has up to 12 cores and connects equipment such as colour light signals, point machines, track circuits, AWS, and level crossing equipment.

Power Cables

Power supply cables are used to carry electrical power to a range of signalling equipment and normally carry 650 volts ac supply. A photograph showing a power cable is at figure 7.

Like lineside cables, power cables can be above ground in concrete troughing or buried. They differ in that they normally have two cores. There are no spares.

Types of cables



Figure 5 . Lineside Cable

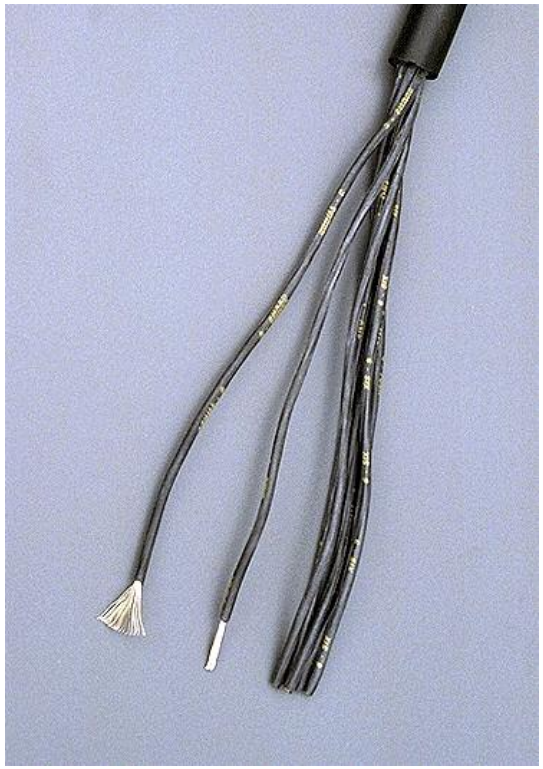


Figure 6. Tail Cable

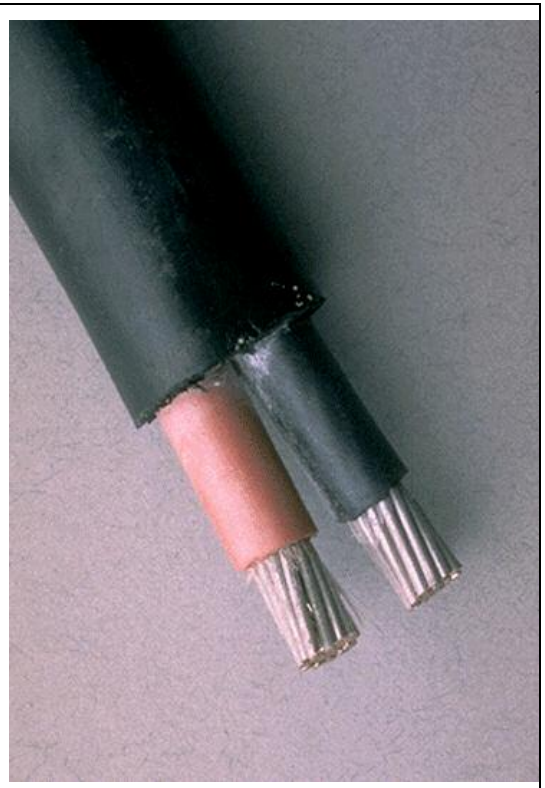


Figure 7. Power Cable

1.4 Assignment 1

You should be able to recognise the cable types you will be working with in the future and distinguish between them. Make a list of the different types of cables used in your work area and where they are used.

Type of Cable	Location of Cable

Visit a location in your work area and note the number of cables feeding into and out of it. Compare your observations with the location diagrams. Ask your mentor to check your list and discuss the information with you.

1.5 Self Assessment Questions

1. What are the two main functions of signalling cables?

2. Name the four components of a cable.

3. How is a large power cable made more flexible and less vulnerable to stress?

4. What damage to the insulation may occur after a short circuit?

5. How is the ‘A’ end of a multicore cable identified?

6. What are the protective properties of both insulation and sheathing?

7. Name the three types of signalling cable.

8. What are spare cores used for ?

9. How many cores are there in a power supply cable?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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2 Visual Inspection of Cables

2.1 Looking for Damaged Cables

Cables are exposed to weather, to vibration, to damage caused by maintenance work and to vandalism. Damage to cables caused by any one of these hazards can result in faults in signalling circuits. Although they are regularly tested to prove their integrity, it is good engineering practice to be on the lookout for signs of physical damage to cables at all times when you are working on or moving along the trackside.

The obvious method of detecting damage is by visual inspection. When damage is identified promptly, repairs can be carried out before a signalling fault occurs. Remember prevention is always the best policy. Be observant at all times. You will need to be able to recognise the different types of cable damage when faulting a circuit.

2.2 Types of Cable Damage

- **Cutting** - This can be the result of a number of things such as chaffing on sharp edges of equipment or ballast. It may be accidental damage by a hand tool or a direct act of vandalism.
- **Crushing** - This may be the result of cable becoming misplaced between or under other equipment but is more likely to be caused by vehicle or mobile equipment wheels.
- **Overheating** - This can be due to internal overheating if a fault is present or from external heating. Damage is most easily recognised by discoloration of insulation or sheathing.

- **Ageing** - Deterioration through age can be recognised by the hardening, cracking or splitting of cable sheath/insulation which visibly exposes inner insulation and conductors.
- **Contamination** - This can be caused by any of a variety of substances coming in contact with the cable and causing the sheath or insulation to deteriorate. Typical contaminants are oils and greases which are commonly used on other railway equipment. You may meet contamination that is potentially dangerous to yourself. This could be in the form of acid spillage from batteries or, in rural areas, agricultural chemicals. Contamination by rodent urine and excreta is a particular health hazard.
- **Rodent damage** - Cables are frequently gnawed by rodents causing significant damage.

Examples of damaged cables are shown in figures 8 to 11.

2.3 Safety Precautions When Examining Cables

Visual inspection of cables as a routine check means '**Look Don't Touch'**. There is no point in disturbing cables unnecessarily as this could do more harm than good.

However, if there is a need to examine specific cables which are known to be faulty, the superficial 'Look Don't Touch' approach must be taken a stage further. By careful handling and closer inspection of cables, loose connections and broken internal conductors can be located. This may involve the removal of troughing lids, opening of location cases and handling cables to effect a complete inspection both above and below each run. It may also involve exposure to hazards to personal safety.

- Exposed wiring may be carrying lethal voltages. Do not handle any cable until you are certain it is disconnected.
- Contamination may be caused by a dangerous substance and it can be invisible to the eye. Do not handle any cables without protective clothing/gloves/goggles.

Examples of Damaged Cables



Figure 8. Cut Cable



Figure 9. Crushed Cable

Examples of Damaged Cables

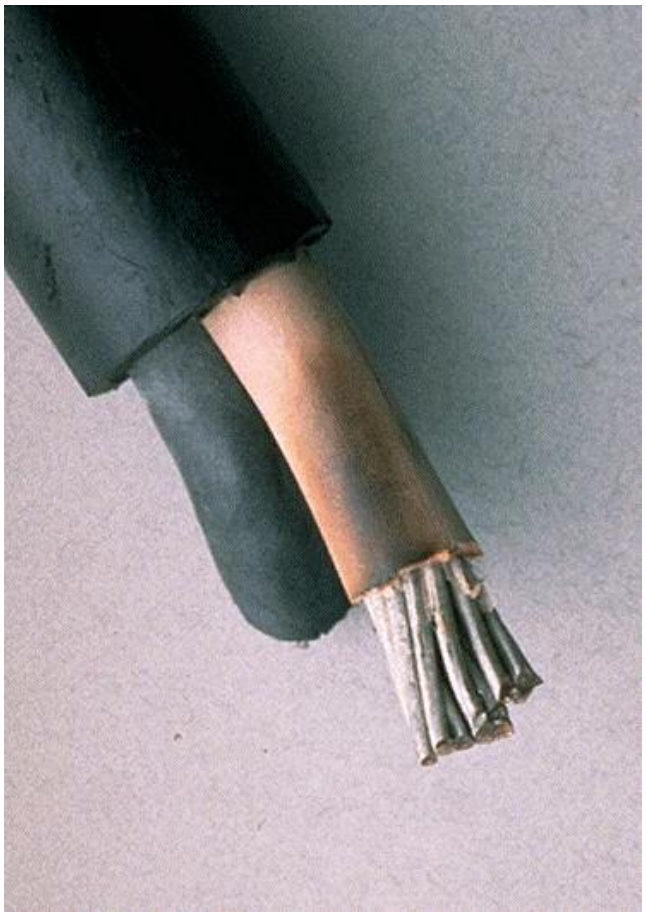


Figure 10. Overheated Insulation



Figure 11. Burned Cable

2.4 Assignment 2

Ask your mentor to show you any damaged cable that has recently been replaced. Talk to your colleagues and find out what kinds of damage are most often encountered in your workplace and where it is most likely to occur

2.5 Self Assessment Questions

1. State two good engineering practices for visually examining cables.

2. What are the five main types of cable damage you can find by visual inspection?

3. State two ways in which damaged cable can be a hazard to your personal safety?

4. What precautions must you take to avoid injury when examining cables?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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3 Visual Inspection of Cables

3.1 Introduction to Testing

Testing cables will enable you to check that they are working correctly. This section explains the testing methods you can use.

Some typical faults are shown in figures 13 and 14.

3.2 Initial and Periodic Testing

A new cable must always be tested when it is installed or fitted as a replacement for an existing cable. To maintain the integrity of cables once they are installed, periodic testing is carried out. The purpose of this testing is to detect potential failures before they affect a vital operational circuit.

There are two significant factors governing the periodic testing of cables. They are:

- **Timing** - The frequency of periodic testing depends on the age of the cable. This varies for different installations and is defined in SMS CA11.
- **Condition**s - The insulation resistance of electrical equipment is affected by the weather. The insulation resistance of a cable is likely to be highest in dry weather and lowest when it is damp. To assess the minimum insulation capability of a cable, testing is always carried out in damp conditions. See SMS CA11 for details of this.

3.3 Testing Methods

There are two methods of testing you will be called upon to carry out on signalling cables. They are

- **Continuity testing** - This is the method of proving that there is a clear and unbroken electrical path from one cable end to the other along each conductor or core of the cable.
- **Insulation Resistance Testing** - This is a test of the resistance to electrical current flow between
  - cores within a cable
  - each core and earth.

3.4 Continuity Testing

Aims of the Test

A continuity test is carried out on each core in a cable to ensue that

- there is no break in the electrical path
- each end of the cable is correctly labelled

Method

It is a simple matter to carry out a continuity test on a cable in the workshop by connecting the two leads from the test meter to the two ends of the cable. However, once the cable is installed the ends will be some considerable distance apart. To overcome this difficulty, we use the following method to carry out the test.

With both ends of the cable disconnected, two cores are selected for testing. These two cores are linked together at one end (see figure 12), and the test meter is then connected across the opposite ends as shown in figure 13. In this way the continuity of both cores is tested at the same time.

The same method is used to test the remaining cores. The simplest way of doing this is to use one core as the reference and to test each of the others with it in turn. After the first test the reference core is known to be sound and any subsequent failure of a test will indicate precisely which core is faulty.

Results

Study the diagram at figure 13 to see how each core is tested. By following the line of test it can be seen that the broken and cross-connected cores will not give a correct continuity reading.

Continuity Test Clips

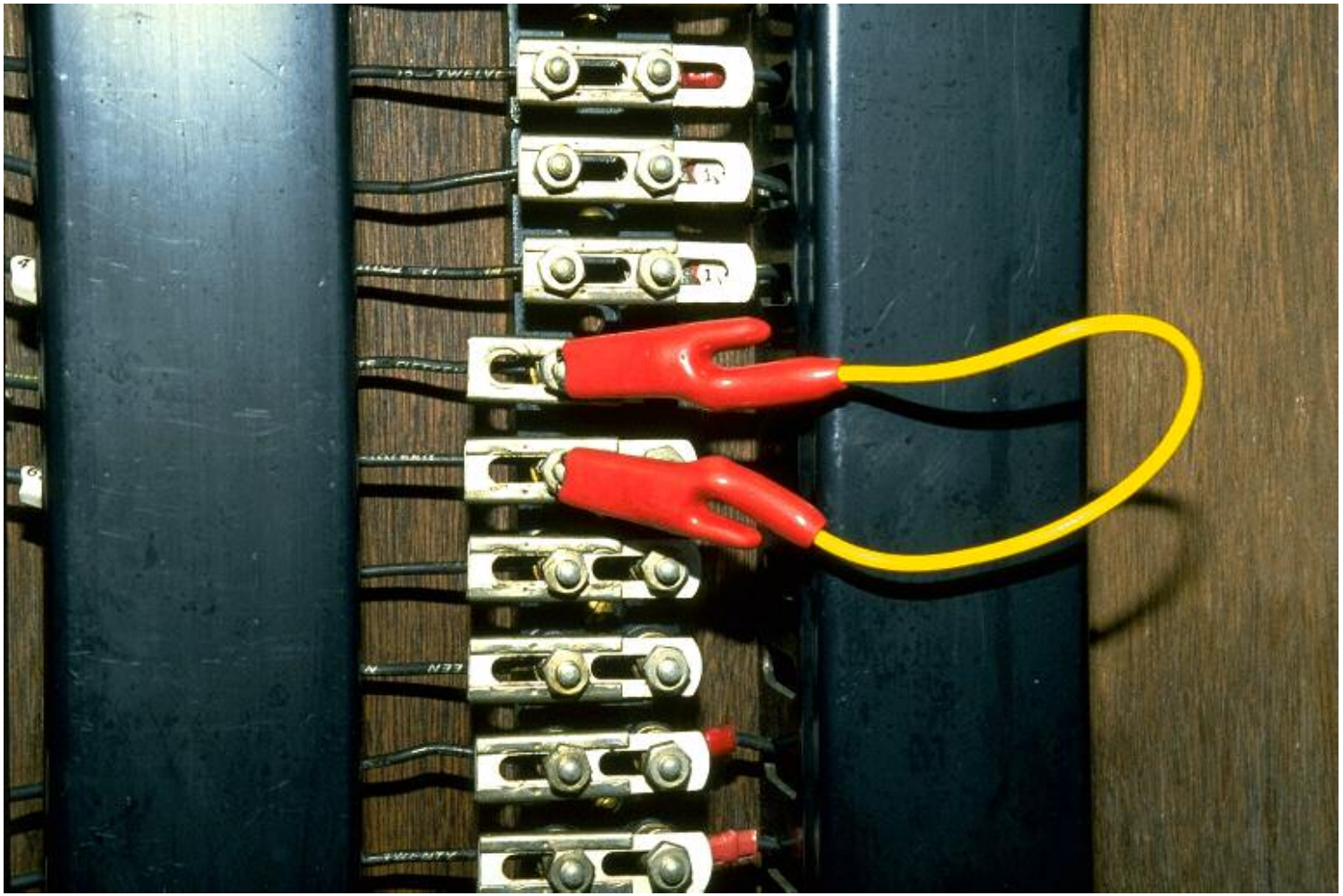


Figure 12

Continuity Testing

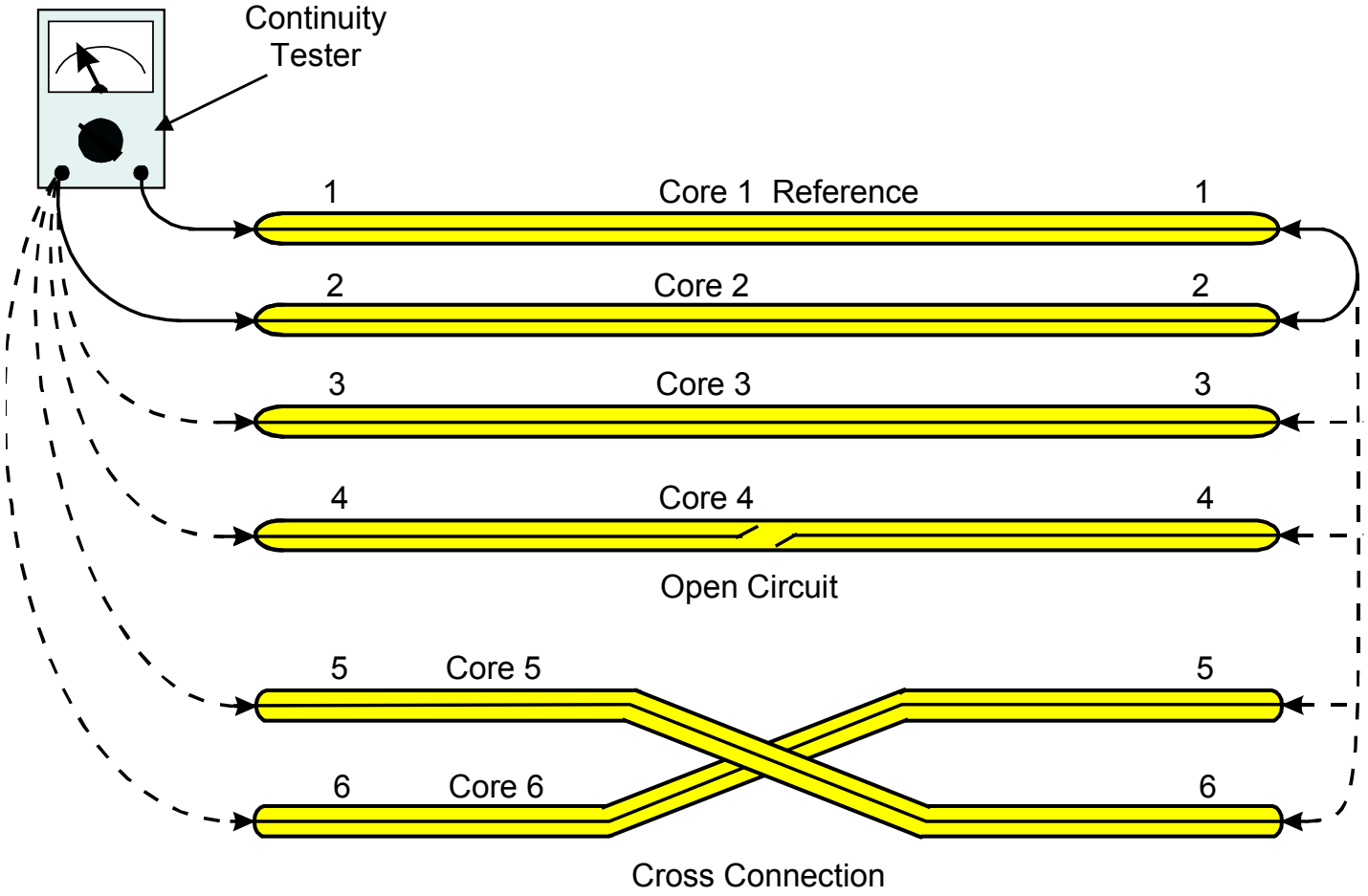


Figure 13

3.5 Insulation Testing

Aims of the Tests

The aim of carrying out insulation testing is to ensure that there is no current leakage from one core to earth, or between cores. Two separate measurements are made of the insulation resistance.

Precautions

Periodic testing of lineside cables is normally only carried out on the spare cores to avoid disconnecting operational equipment. It is assumed that, if the spare cores which are in the outer, more vulnerable, layer of the cable are electrically sound then the remainder will also be sound.

In the event of other cores requiring test, or if the test is carried out on a power cable which has no spare cores, both ends of the cable must be disconnected before testing begins.

Methods

- **Core to Earth Test** - In preparation for the test it is necessary to establish a good reference earth point. If there is an earth point on an adjacent equipment, it can be used. If not, then an earth must be established by making a connection to a suitable piece of metal framework or using an earth spike driven into the ground. Either can be verified using the insulation tester prior to testing the cable. The core to earth test is a series of individual tests. One lead from the test meter is connected to the reference earth point and the other to each core in turn.
- **Core to Core Test** - Each core is tested to all the other cores connected together. Approved clips are used to make the temporary joint of the cores and have to be accounted for after the testing is completed.

Insulation Faults

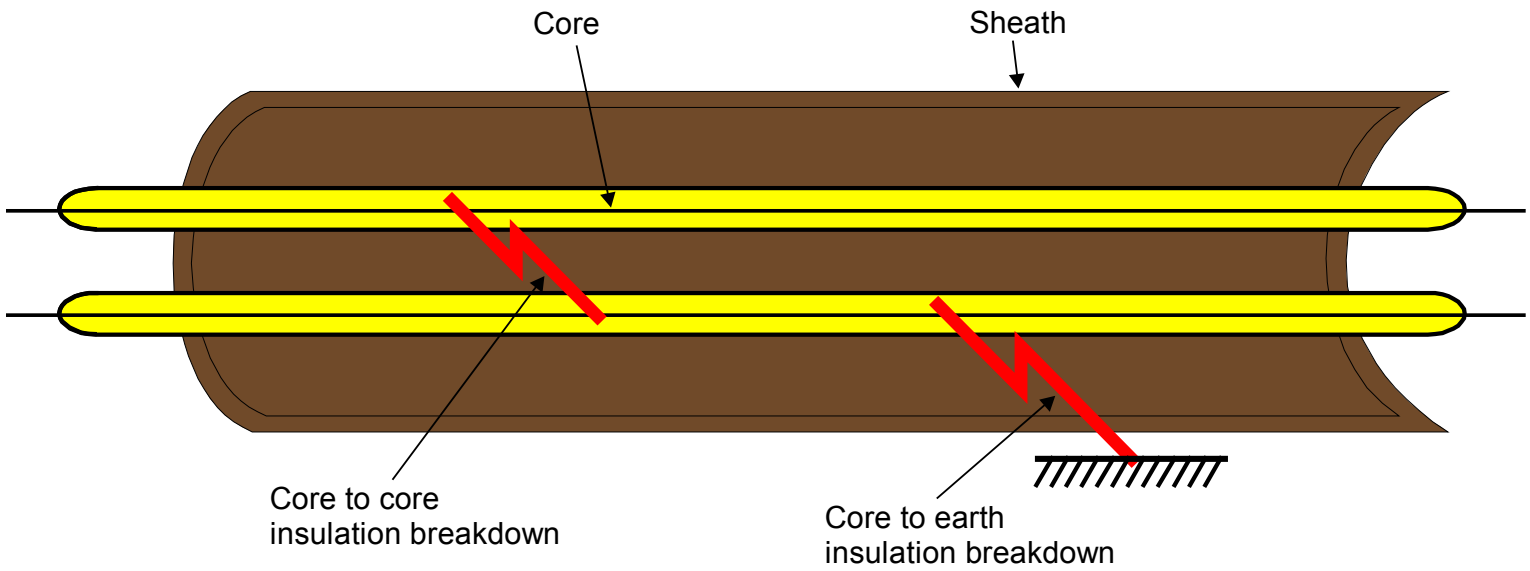


Figure 15

3.6 Assignment 3

It is important that you understand the logic behind the cable tests you carry out.

Ask your mentor to let you observe a periodic cable test. Identify the circuits that are being tested and relate the tests to the circuit diagram. Ask you colleagues to explain the tests as they are carried out.

Do not worry about the use of individual test meters. You will be instructed on their use in another module.

3.7 Self Assessment Questions

1. What are the two methods of testing to prove that cables are serviceable?

2. What types of fault will each test identify?

3. What are the two factors that determine when periodic testing will be carried out?

4. How would you carry out a continuity test on a lineside cable which is 50 metres in length?

5. What are the two types of insulation test that are carried out on lineside cables?

6. What would you do before testing a power cable?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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4 Safety Precautions

4.1 Introduction

There are two aspects to safety when working with signalling cables. The first is the need for personal safety and the second is the need to ensure the safety of equipment.

4.2 Personal Safety

Your personal safety will be maintained if you

- Disconnect electrical power from all cables before you work on them. This applies equally to testing and to replacing cables.
- Only handle cables when wearing protective clothing, including gloves and goggles. Some possible contamination, such as battery acid, agricultural chemicals and rat urine, may not be easily visible.

4.3 Equipment Safety

It is just as important to ensure the safety of signalling equipment and as it is to protect yourself. A responsible approach is essential at all times when working with signalling cables to ensure the safety of equipment. This in turn will protect passengers and staff over a wide area. To ensure equipment safety you must

- never disconnect operational cables without the authority of your supervisor and/or the signalperson
- never make unauthorised alterations to signalling cable connections or routing

4.4 Assignment

Find out what safety equipment is held for use at your depot and make sure you know how to obtain it when it is needed.

Check your local instructions for safety. Make sure you know what the regulations are for handling and working on cables.

4.5 Self Assessment Questions

1. State the two main precautions you must take when handling or working with signalling cables.

2. Whose authority do you need before disconnecting operational signalling cables?

3. When can you make alterations to signalling cables?

If you are not totally confident that you have correctly answered any of these questions, re-read the relevant parts of this section to check your answer(s) before going on to the next section.

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5 Summary

5.1 Introduction

Signalling cables are used to carry commands from the interlocking to trackside equipment and, in the opposite direction, indications of trackside equipment state to the interlocking. There are four component parts to cables; conductor, core, insulation and sheath. There are three main types of signalling cables; lineside, tail and power.

5.2 Visual Inspection of Cables

The timing and methods of visual inspection of cables is important because they are vulnerable to several forms of damage. The types of cable damage that are most likely to be found have been described. Your attention is drawn to the safety precautions to be observed when working with signalling cables.

5.3 Cable Testing

Initial/periodic testing of signalling cables is carried out in two ways. Continuity testing proves the integrity of cores and that they are correctly terminated. Insulation testing is carried out between cores and from core to earth. These two methods of testing reveal a range of faults which can cause failure of the communication with trackside equipment or the power supplied to it.

6 Self-Test

6.1 Introduction

This section contains a test by which you can test your knowledge of the objectives for the Principles of Signalling Cables. The test is provided to give you experience of the type and difficulty of the questions you will be given in the Module 6 assessment.

The self-test contains questions which require the following types of answer:

- **True/False**. The answer will be either true or false. Circle the correct answer.
- **Multi-choice**. You will be offered three answers, only one of which is correct. Circle the identifying letter of the correct answer.
- **Short Answer**. You will need to write down the answer in the space provided.

You should attempt every question. The answers are given in Section 7.

6.2 Questions

Question 1 One of the functions of a signalling cable is to carry electrical power to signalling circuit components.

True/false

Question 2. Cables can be damaged when maintenance work is carried out. Write down three other hazards to which cables are exposed.
1.
2.
3.

Question 3. The conductor of a cable is usually made of:

a. Steel

b. Copper

c. Bronze

Question 4. The end of a multi-core cable, where the numbering of cores increases clockwise from the centre is called the:

a. ‘A’ end

b. ‘L’ end

c. ‘Z’ end

Question 5. The purpose of the sheath in a cable is to:

a. Electrically separate conductors in the cable

b. Protect the cable from damage

c. Provide a good electrical conductor

Question 6. Spare conductors are provided in a lineside cable to:

a. Allow connection of additional equipment at a later date

b. Increase the strength of the cable

c. Reduce the electrical resistance of the cable

Question 7. Tail cables are usually buried in the ground.

True/false

Question 8. The AC voltage carried by a power cable is normally:

a. 60 volts

b. 230 volts

c. 650 volts

Question 9. Hardening, cracking or splitting of a cable's sheath or insulation is a sign of:

a. Ageing

b. Crushing

c. Cutting

Question 10. Protective clothing must be worn when handling a cable.

True/false

Question 11. The principle to be followed when visually inspecting cables is to:

a. Touch and feel

b. Look and feel

c. Look don't touch

Question 12. A new cable must always be tested when it is fitted as a replacement for an existing cable.

True/false

Question 13. Periodic cable testing is always carried out when climatic conditions are:

a. Dry

b. Damp

c. Wet

Question 14. An open circuit in a conductor will be found by:

a. Insulation testing

b. Earth testing

c. Continuity testing

Question 15. Periodic testing of lineside cables is normally only carried out by disconnecting operational equipment.

True/false

Question 16. When carrying out a core-to-core test, the cores are temporarily connected together by:

a. Using approved clips

b. Twisting cores together

c. Connecting a shorting plug

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7 Answers

7.1 Answers to Self-Test Questions

Question 1. True	Question 9. a
Question 2. Vandalism, weather, vibration	Question 10. True
Question 3. b	Question 11. c
Question 4. a	Question 12. True
Question 5. b	Question 13. b
Question 6 a	Question 14. c
Question 7. False	Question 15. False
Question 8. c	Question 16. a

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8 Glossary

cable - used to connect together electrically the components in a signalling circuit.

command - communication from the interlocking to the trackside equipment.

communications - the passing of commands and indications.

conductor - a carrier of electric current, it is usually made of copper.

continuity testing - testing that a conductor has an unbroken electrical path from one end of a cable to the other.

core - a single conductor in a group of conductors but insulated from all others.

indication - communication of equipment state from the lineside to the interlocking.

initial testing - the testing of a cable when it is first installed or fitted as a replacement for an existing cable.

insulation - the part of a cable which keeps each conductor electrically separate from other conductors in the cable.

insulation testing - testing the electrical resistance between cores and between a core and earth.

lineside cable - a cable running from the interlocking to location cases and between cases.

periodic testing - the testing of a cable at periodic intervals to detect potential failures.

power cable - a cable used to supply electrical power to a signalling equipment. It normally carries 650 volts AC.

sheath - the outer layer of a cable designed to protect the cable interior cores from damage and deterioration.

tail cable - a cable running from a location case to trackside equipment.